

Abdul Rahman Mohamed Farharth

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Professional Summary

I am a third-year biomedical engineering undergraduate with interests in medical devices, medical imaging, biomedical signal processing, and healthcare technology development. Experienced in embedded systems, physiological signal analysis, medical image processing, and hardware-software integration for healthcare applications. Passionate about developing and validating safe, reliable medical technologies.

Research Interests: Medical Devices, Medical Imaging, Biomedical Signal Processing, Ultrasound Imaging, Embedded Medical Systems, Medical Device Verification, Healthcare Technology Development

Education

University of Moratuwa, Sri Lanka

2024 – Present

B.Sc. Engineering (Hons.) in Biomedical Engineering

Expected Graduation: 2028

Department of Electronic and Telecommunication Engineering

- CGPA: 3.79/4.00 (First Class Honours)
- Dean's List – Semesters 1 & 2
- Coursework: Biomedical Instrumentation, Biomedical Device Design, Electronic Circuit Design, Electronic Design Realization, Electronic Control Systems, Signals and Systems, Digital Signal Processing, Image Processing and Machine Vision, Medical Imaging, Pattern Recognition, Modelling and Analysis of Physiological Systems, Data Structures and Algorithms, Internet of Things

Zahira College, Colombo, Sri Lanka

2020 – 2023

G.C.E. Advanced Level Examination

- 4As Distinction Passes (Combined Mathematics, Physics, Chemistry, General English)
- Island Rank: 407 / Z-score: 2.3330

Technical Skills

- **Programming:** Python, MATLAB, C, C++, R (Basic)
- **Data Analysis:** NumPy, Pandas, Matplotlib
- **Machine Learning & Image Analysis:** PyTorch, OpenCV, Ultralytics YOLO, Faster R-CNN, DenseNet-121, EasyOCR
- **Signal Processing:** FFT, Digital Filtering, Physiological Signal Processing, Feature Extraction, ECG, EMG, EEG, Data Acquisition Systems
- **Biomedical Engineering:** Medical Imaging, Medical Device Prototyping, Physiological Signal Acquisition
- **Testing & Validation:** System Integration, Hardware Debugging, Prototype Testing, Sensor Calibration, Verification Documentation
- **Hardware:** Arduino, ESP32, ESP8266, Raspberry Pi 5, ATmega328P
- **Software:** SOLIDWORKS, Altium Designer, LTSpice
- **Laboratory Skills:** Oscilloscope, Function Generator, Digital Multimeter, Soldering, PCB Assembly, Circuit Debugging

Projects

Temperature-Controlled Vaccine Cold Chain Monitoring System 🐙

Semester 3 Project

- Designed, assembled, integrated, and verified a portable biomedical cold-chain monitoring system using ESP32/ESP8266 microcontrollers, wireless communication, and temperature sensing. Performed hardware testing, sensor calibration, and system-level validation to ensure reliable operation.

Smart Posture Monitoring Cushion

Semester 2 Project

- Developed a smart posture-monitoring cushion using 8 FSR402 pressure sensors and ultrasonic backrest sensing to detect poor sitting posture in real time, providing haptic feedback to alert users during slouching; achieved nearly 70% slouch detection accuracy.
- Designed the PCB enclosure, battery housing, and sensor mounting structure in SOLIDWORKS. Contributed to hardware prototyping, sensor placement testing, circuit assembly, and overall system integration. Evaluated different sensor layouts and feedback strategies to improve comfort and usability during prolonged sitting.
- Performed sensor characterization, hardware integration, and functional testing to optimize system reliability and user performance.
- Awarded 2nd Runners-Up at Brainstorm 2025 (University of Moratuwa engineering competition) among university-wide healthcare innovative engineering projects.

Glove Rejection and Inspection Process (GRIP)

Ongoing - Semester 4 Project

- Developing a real-time machine vision inspection system integrating embedded computing, computer vision, robotic actuation, and hardware-software verification for automated industrial inspection.
- Investigating human-in-the-loop vs. fully automated decision-making frameworks for industrial human-robot interaction.

Thyroid Nodule Detection and Malignancy Classification

Independent Project - Kaggle

- Developed a deep learning-based medical imaging pipeline for thyroid nodule detection and benign/malignant classification from ultrasound images using Python and PyTorch. Performed image preprocessing, model training, performance evaluation, and deployment through an interactive Gradio web application.

Certificates

- **Machine Learning Specialization** – Deep Learning Specialization (Coursera) 2025
- **Medical Image Processing (Biomedical Imaging & Analysis)** – MathWorks (MATLAB) 2025

Achievements

- Awarded 2nd Runners-Up at Brainstorm 2025 (University of Moratuwa engineering competition)
- Dean's List - Semesters 1 & 2, University of Moratuwa
- Awarded Principal's Gold Medal for Best Performance in the G.C.E. A/L Physical Science Stream
- Sri Lanka Mathematics Olympiad Competition 2018 - Distinction Award

Leadership and Volunteering

- IEEE Engineering in Medicine and Biology Society (EMBS), University of Moratuwa
 - Finance Lead (Brainstorm 2026), Social Media Manager
- IEEE Signal Processing Society (SPS), University of Moratuwa
 - Event Chair
- Department Representative, Department of Electronic and Telecommunication Engineering

References

Dr. Ranga Rodrigo

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